

NEUTRON DOSE RATE METER 2202 D

The AE 2202 D Neutron Dose Rate Meter is a portable instrument designed for radiation protection survey of areas around reactors, accelerators, neutron sources or other neutron producing equipments. The instrument measures the neutron dose rate in the unit of mrem/h with approximately correct rem dose response in the energy range from thermal to 17 MeV.

AE 2202 D is a complete measuring assembly consisting of a neutron detector unit and an electronic unit mounted together.

Detector unit

The detector unit consists of a special BF_3 proportional counter tube of high neutron sensitivity surrounded by a shield made of polyethylene and boron plastic (1). A new type of BF_3 counter is used where the anode wire support needs less space than what is normally required. In this way it has been possible to reduce the total length of the detector unit without deteriorate the rem dose response. This modification together with a slightly higher pressure in the counter has made it possible to improve most of the characteristic data for the instrument. The shield has a moderating and absorbing effect on the neutrons, and it is arranged such that the overall neutron sensitivity of the unit is dependent on neutron energy in the same way as the biological damage effect on humans from neutrons. The instrument can thus be used to measure the dose rate in mrem/h without the necessity to know the neutron spectrum. Fig. 1 shows the construction of the detector unit. The thickness of the two polyethylene layers and the hole area in the boron plastic layer have been chosen to obtain the rem response.

Fig. 2 shows the energy response measured for the detector together with the curve given by ICRP as the ideal rem dose response. Close fit to the ICRP curve has been achieved over an energy range from 0.025 eV - 17 MeV. The Neutron Dose Rate Meter is essentially non-directional in response from $\theta = 0$ to 90° as shown in Fig. 3. The 90° direction should preferably be used. The deviation from an ideal rem response of 3.6 cps per mrem/h is shown in Fig. 4.

(1) I.Ö. Andersson, J. Braun: A Neutron Rem Counter, Nukleonik 6.
5 Heft p.237 (1964)

For comparison, finally, the neutron dose in rem per n/cm^2 as a function of neutron energy is shown in Fig. 5. The curve is taken from NBS Handbook 63 (1957).

The use of a BF_3 counter for neutron detection offers the advantage of very efficient discrimination against gamma background, and with correct bias setting the AE 2202 D is insensitive to gamma radiation up to 200 R/h.

The BF_3 counter is operating at a voltage of +2200 V. This voltage has been chosen to acquire max. discrimination against gamma background. See Fig. 6. The detector sensitivity is such that a pulse rate of 3.6 cps corresponds to a neutron dose rate of 1 mrem/h. With the discriminator level at 2.0 V the contribution from 200 R/h gamma background will be less than 0.5 mrem/h.

Measuring range

Continuous neutron radiation can be measured with the AE 2202 D instrument over a wide range. The ratemeter range is from 0.1 to 10000 mrem/h. A pulse output is available for external counting. A recorder can also be connected to the instrument.

Pulsed neutron radiation

If the instrument has to be used for measurements of neutron radiation from pulsed neutron generators it is necessary to know both pulse length and repetition frequency in order to be able to calculate the percentage counting loss for a certain radiation level.

However, since the detector is surrounded by a mass of neutron moderating material, a finite time is required for the slowing down and diffusion of the neutrons. This results in an expanded time distribution of about 60 μs of the neutron burst as seen by the detector. The AE 2202 D is therefore possible to use even for pulsed radiation, but with some limitation specially at low repetition frequencies.

Since each burst will have an equivalent length of about 60 μ s the detector will see almost a continuous neutron radiation as soon as the frequency is higher than $\sim 2 \times 10^4$ Hz. This stretched burst has a maximum rate of neutrons between 10 - 20 μ s after the start of the pulse and the time distribution is independent of the pulse width. (See ref. 2)

If N is the true reading in c/s, and N_0 is the measured counts in c/s we get the counting loss from the formular

$$N = \frac{N_0}{1 - \tau N_0}$$

The result with $\tau = 0.5 \mu$ s is shown in Fig. 7. Curve 5.

If the burst repetition frequency, f, is lower than 2×10^4 Hz and the length is short (i.e. less than 10 - 20 μ s) the true reading will be

$$f \cdot 60 \cdot 10^{-6} \cdot \frac{N_0}{1 - \tau N_0}$$

The percentage counting loss for pulse repetition frequency of f = 1 Hz, 10 Hz, 100 Hz, 1000 Hz in Fig. 7. Curve 1, 2, 3 and 4.

All curves in Fig. 7 are given under the assumption that the influence of gamma background is negligible (i.e. less than 100 - 200 R/h).

Electronic unit

A block diagram for the electronic unit is shown in Fig. 8, circuit diagrams in Figs. 9 and 10 and a wiring diagram in Fig. 11.

The pulses from the detector are amplified in the charge amplifier. Noise and γ -pulses are rejected in the discriminator which gives a standard pulse out for each "neutron pulse" in. The standard pulses are fed to a diode pump. This gives charge pulses, which are integrated in a D.C. logarithmic amplifier with a time constant inverseley proportional to the pulse rate. The standard pulses

are available on the ratemeter board terminal "1" for external counting or on the front of 2202 D. The output signal from the logarithmic amplifier is fed to a meter showing the dose rate in mrem/h and to terminal "5" on the ratemeter board.

The H.V. supply for the detector consists of a stabilized voltage supply, a D.C.-A.C. converter, a high ratio voltage transformer and a cascade diode rectifier. See Fig. 9. With the switch S3 located on the chassis, the high voltage can be changed in three steps: 2100 V, 2200 V and 2300 V in order to verify that the countrate-H.V. plateau is correct. See Fig. 12.

Range (at uniform dose rate)

Log. ratemeter:

0.1 - 10000 mrem/h

Neutron sensitivity:

3.6 cps/mrem/h

Gamma sensitivity:

Insensitive up to 200 R/h

Min. pulse charge from the detector:
(high voltage = 2200 V)

$5 \cdot 10^{-14}$ Coulombs

Charge amplifier

Amplification:

$2 \text{ V}/10^{-13}$ Coulombs

Differentiation time:

1 μ s

Integration time:

Inversely proportional to the dose rate with 3 s at 2.5 mrem/h with the switch "TC" in position "SHORT" and 6 s in position "LONG".

Discriminator

Level range:

1.3 - 2.3 V

Normal level set point:

2.0 V

Output pulse for log. amplifier
and ext. counting

Amplitude:

-3.5 V

Duration:

1 μ s

Recorder output:

0 - 1 mA

Power supply

Internal:

Batteries 9 x 1.5 V IEA x20

Operational time:

About 100 h/set

External:

See Fig. 11

H.V. supply

Voltage:

+2200 V

Drift:

< 1 V/°C

Frequency of square wave from
the D.C.-A.C. converter:

~ 10 kHz

S3 in position 1:

2100 V

S3 in position 2:

2200 V

S3 in position 3:

2300 V

Temperature range:

-10°C to +40°C

Dimensions

Diameter:

215 mm

Length:

325 mm

Weight:

10.9 kg, incl. batteries

MAINTENANCE

Before delivery the AE 2202 D is tested and adjusted to correct performance. Information about test and trimming procedure is given here in order to make it possible to verify operation.

Battery check

Push "Battery check" button on the front. The pointer should be within the green area. The main switch must be set in position "ON".

H.V. supply check

The high voltage can be measured at terminal "8" on the ratemeter board (see Fig.10). To get a correct reading use a voltmeter with an input impedance $\geq 1000 \text{ M}\Omega$ or an electrostatic voltmeter.

Amplifier check

A test pulse signal with neg. polarity should be applied to terminal "7" on the ratemeter board via a capacitor of 1 pF. The charge applied to the input is thus $V \cdot 10^{-12}$ Coulombs where V is the amplitude of the test pulse expressed in volts. See Fig. 13.

Discriminator check

The output pulse (see Fig. 14) is checked at terminal "1" on the ratemeter board or on the front of 2202 D. The trigger level can be adjusted with R16 (see Fig. 10).

D.C. logarithmic amplifier check

Feed negative pulses to the amplifier as described under "Amplifier check" and set the level of the discriminator so that it is triggered. Let the pulse

repetition rate be 3.6 pps. Adjust R33 (see Fig.10) so that the meter reading is 1 mrem/h. Change the pulse rate to 3600 pps and adjust R34 so the meter indicates 10^3 mrem/h. Change the pulse rate between 3.6 and 3600 pps and adjust simultaneously R33 and R34 so that the change in meter indication is three decades. Test the meter indication also for 0.1, 10, 10^2 and 10^4 mrem/h.

Total performance check

A neutron source (PuBe, RaBe) is placed to the detector. The charge pulse from the detector should give voltage pulses at the output of the amplifier ranging from 1V to 3V.

For correct gamma pulse discrimination the discriminator level should be set as high as possible without losing any "neutron pulses".

For accurate setting use the external counting to determine the discriminator curve as shown in Fig. 6.

Observe:

If the neutron dose rate sensitivity is controlled with a neutron source of known strength, one must have in mind that the contribution from scattered neutrons may be considerable if not very careful arrangements are made to avoid these.

Otherwise meter readings of 25% or more above the theoretical dose rate may occur.

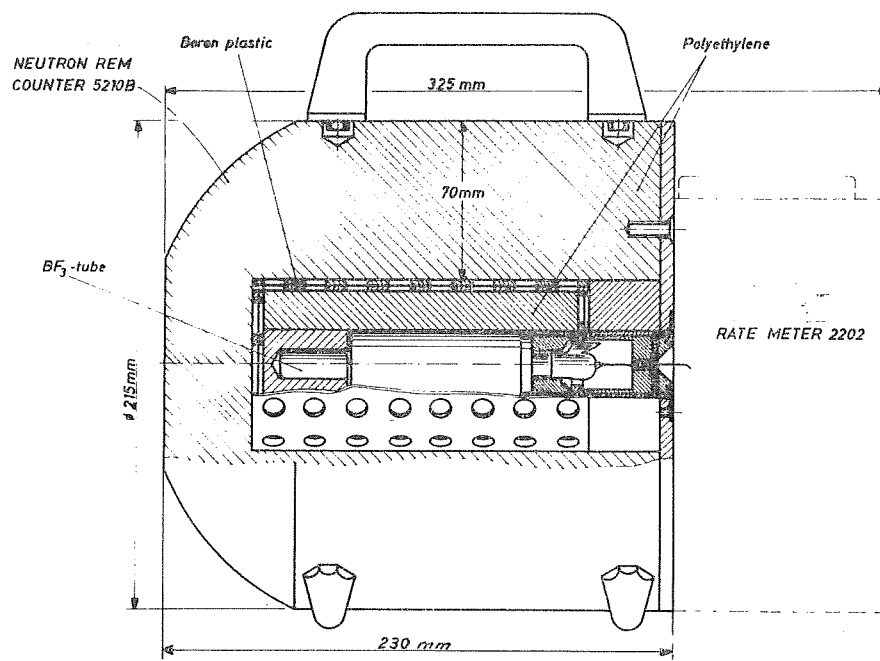


Fig. 1 Neutron Dose Rate Meter 2202 D

- o Broad beam $\theta = 90^\circ$ (See Fig. 3)
- x Broad beam $\theta = 0^\circ$ (- " -)
- Isotropic neutrons

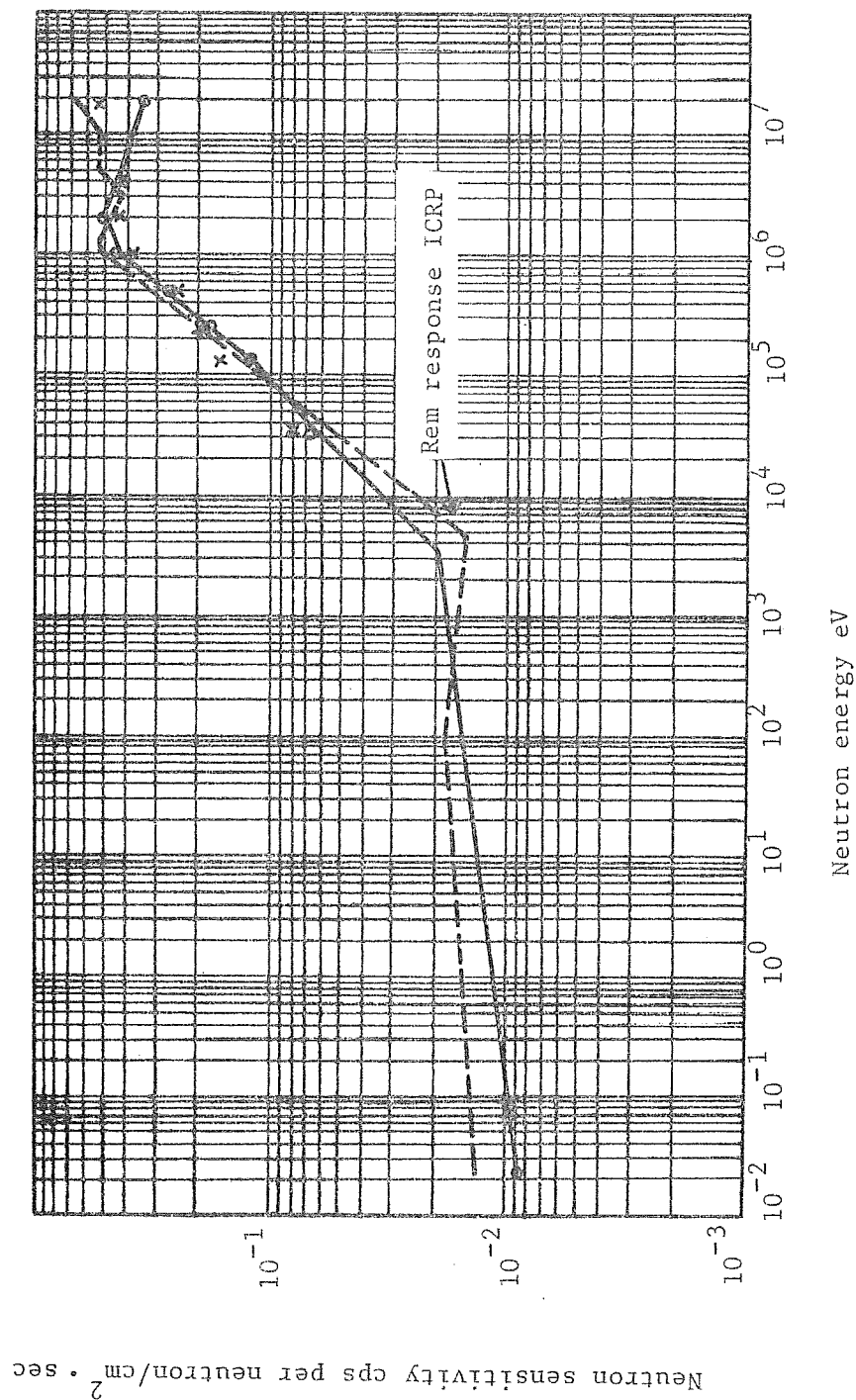


Fig. 2 Neutron sensitivity as a function of energy for Neutron Dose Rate Meter 2202 D

Energy MeV	Direction θ		
	0°	90°	180°
0.126	1.05	1.0	-
0.228	1.05	1.0	0.50
0.516	0.95	1.0	-
1.046	0.90	1.0	-
2.02	0.90	1.0	0.60
4.2	1.00	1.0	0.60
17.1	1.10	1.0	0.50

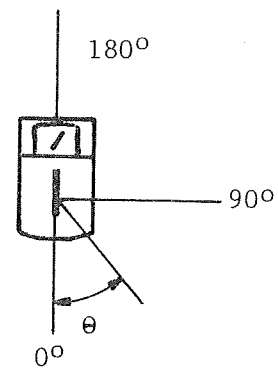


Fig. 3 Angular response. Relative counting rate

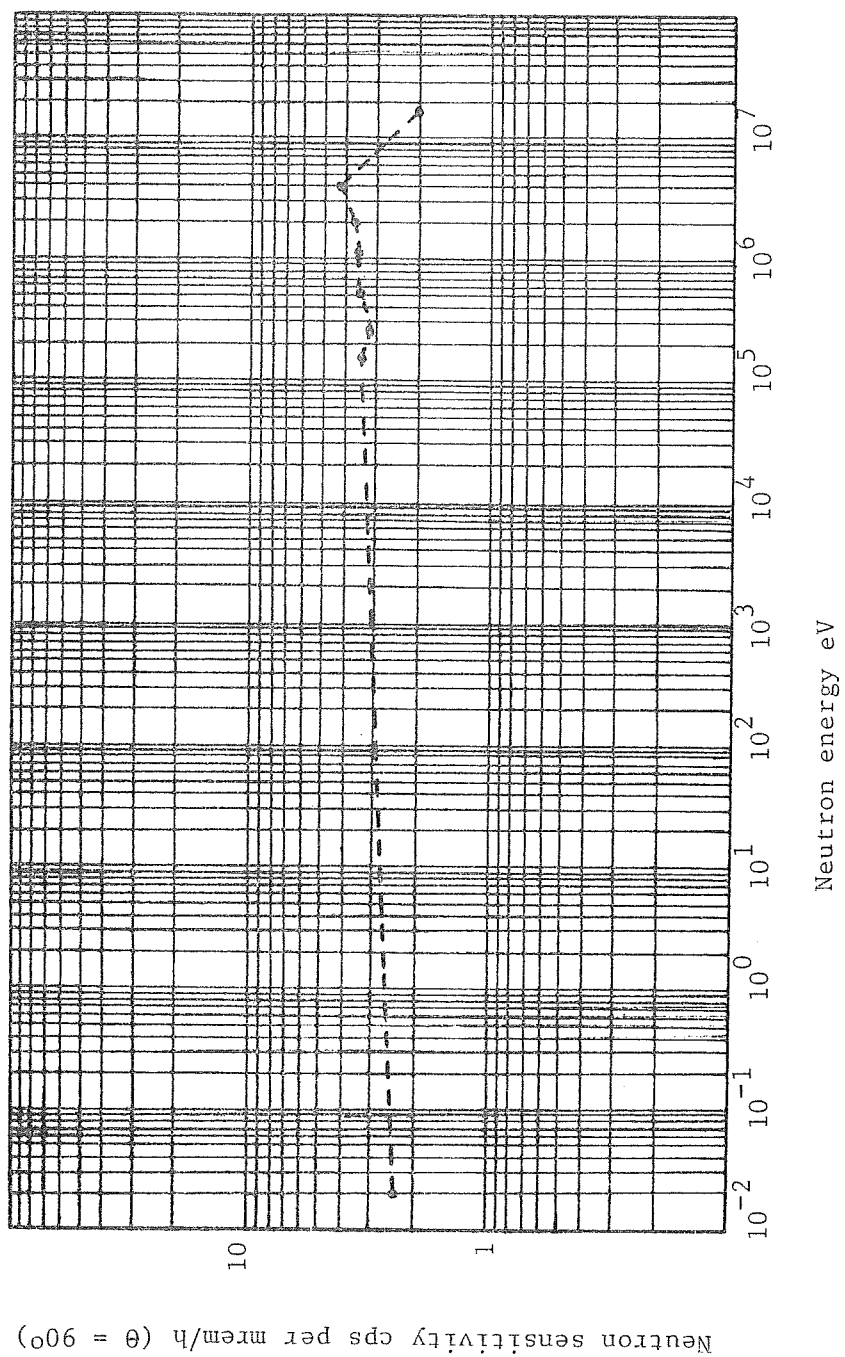


Fig. 4 Response of Neutron Dose Rate Meter 2202 D

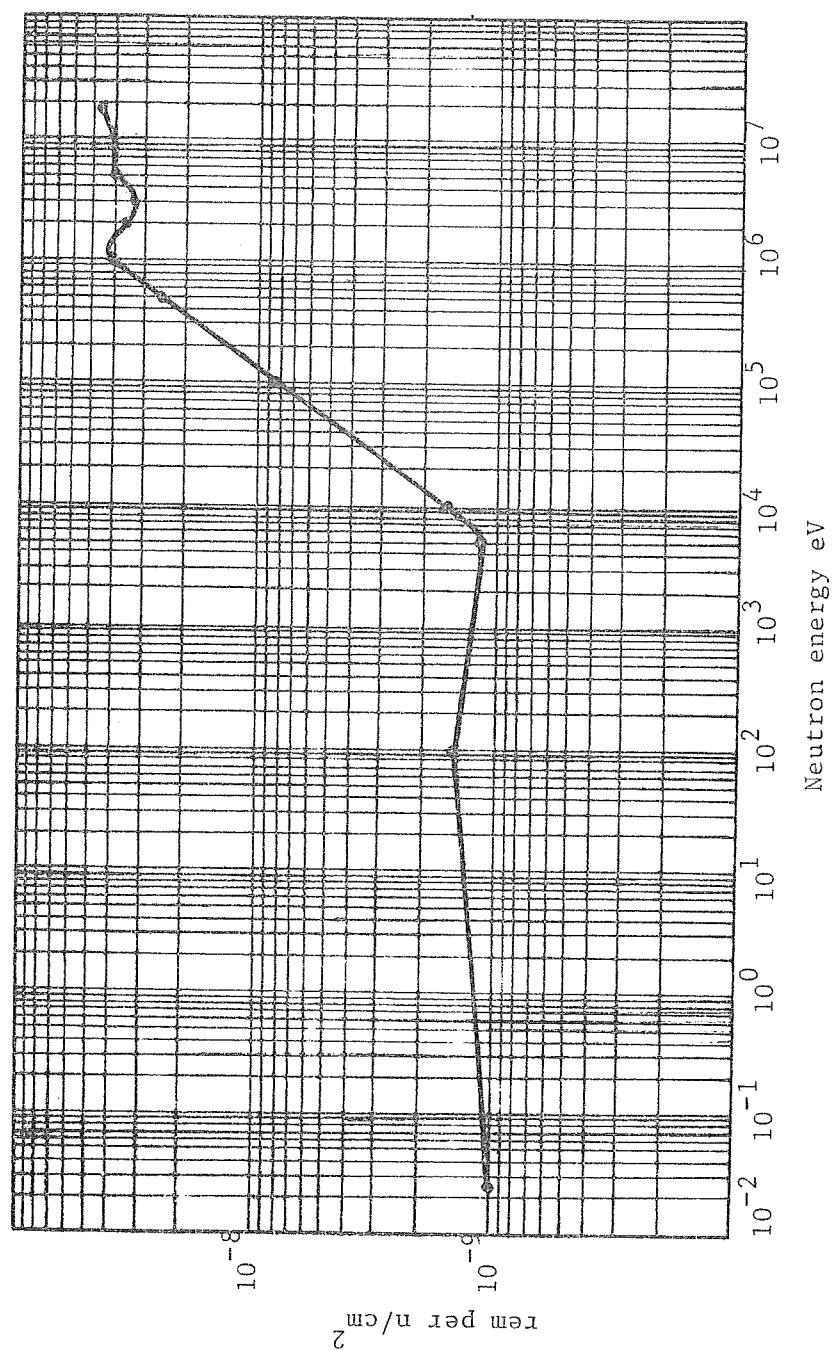


Fig. 5 Neutron dose in rem per n/cm² as a function of neutron energy

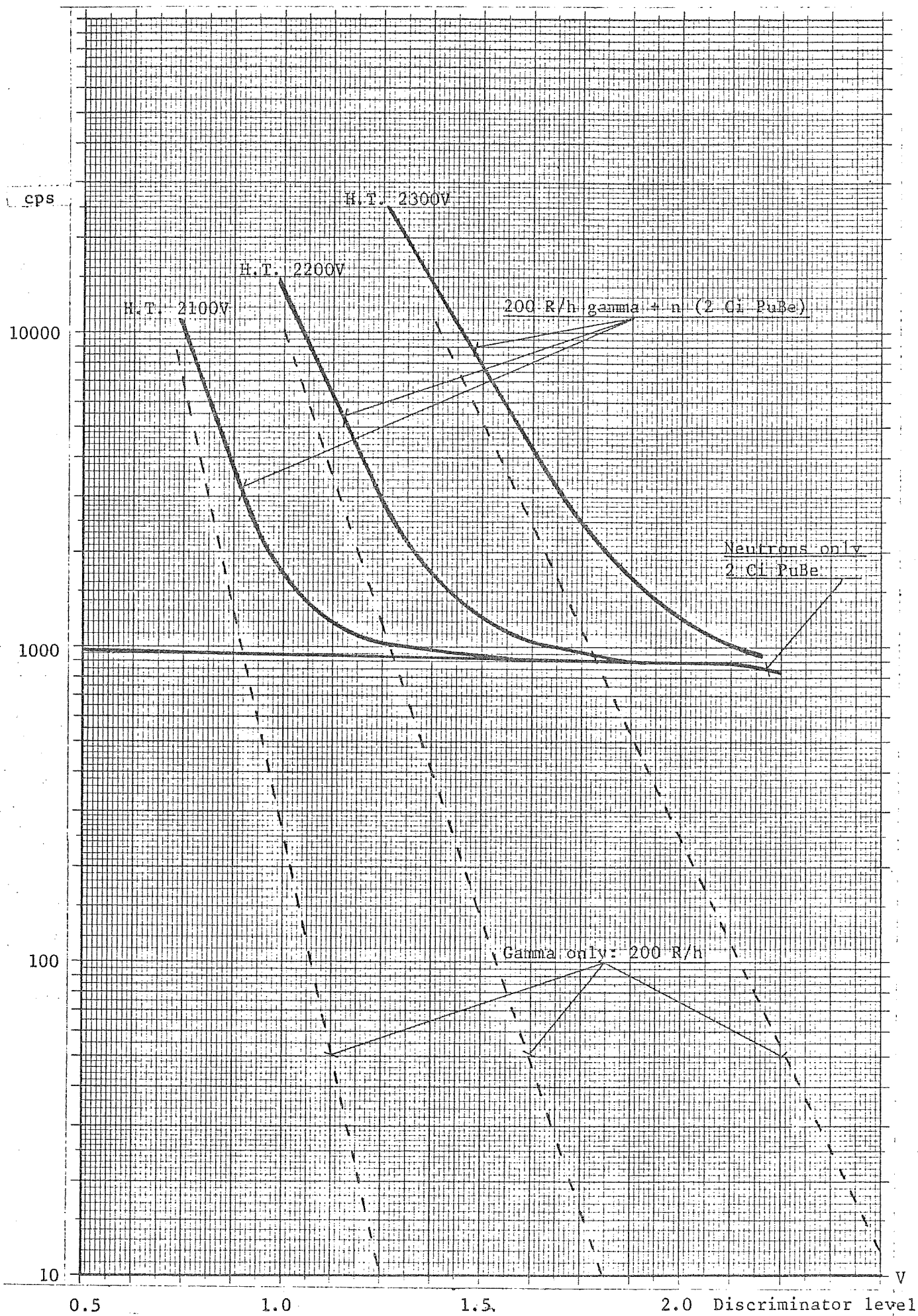


Fig. 6 Gamma background influence

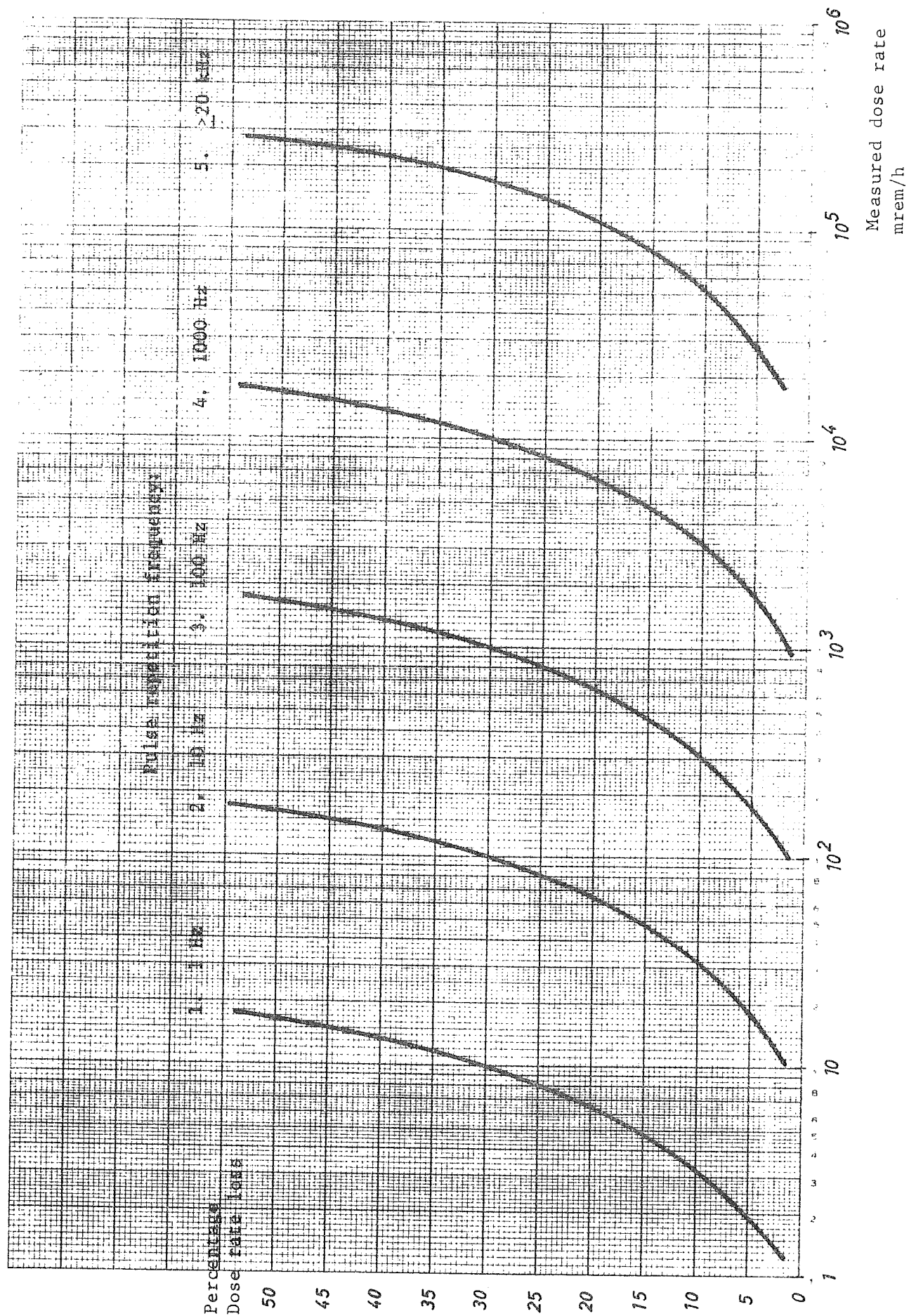


Fig. 7 Counting loss at various pulse repetition frequencies.

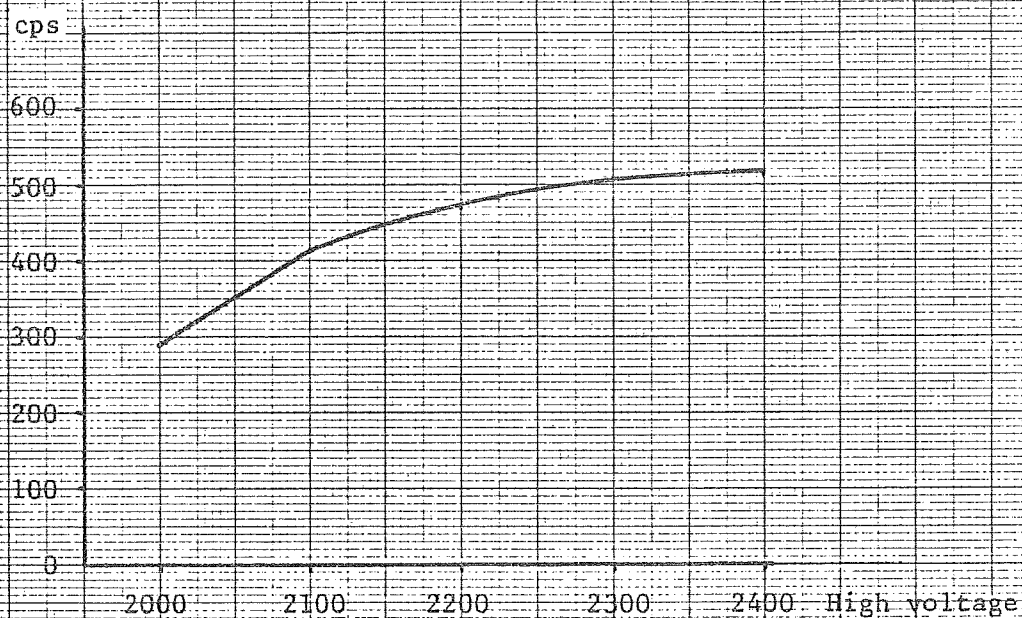


Fig. 12 Typical plateau curve.
1 Ci PuBe at a distance of about
130 mm from center of the BF_3 tube
Instrumentation: 2202 D

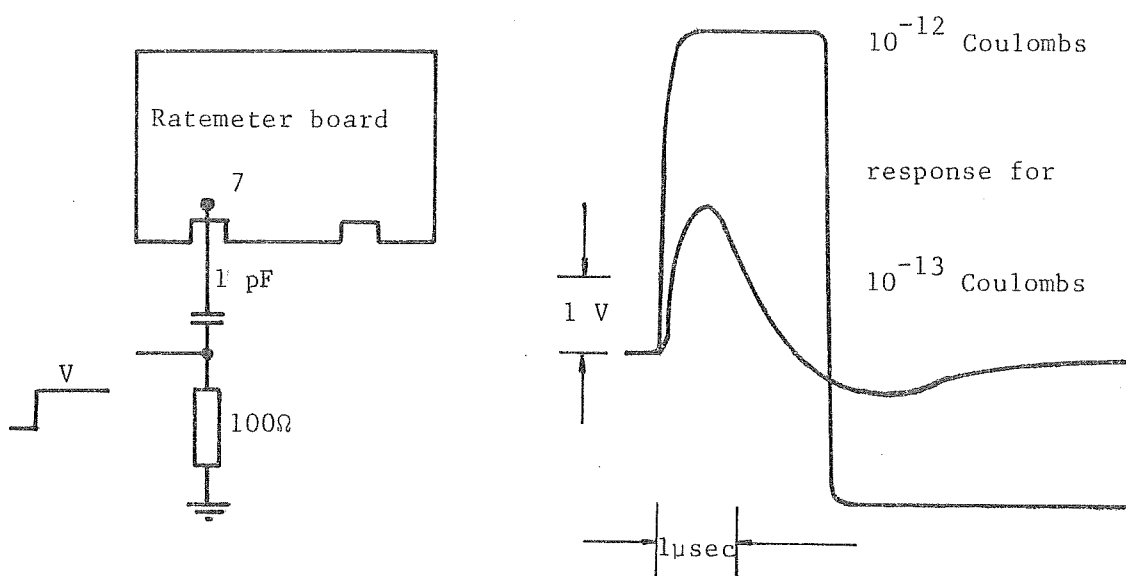


Fig. 13 Amplifier check

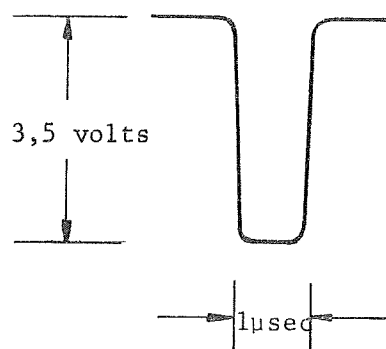


Fig. 14 Discriminator check

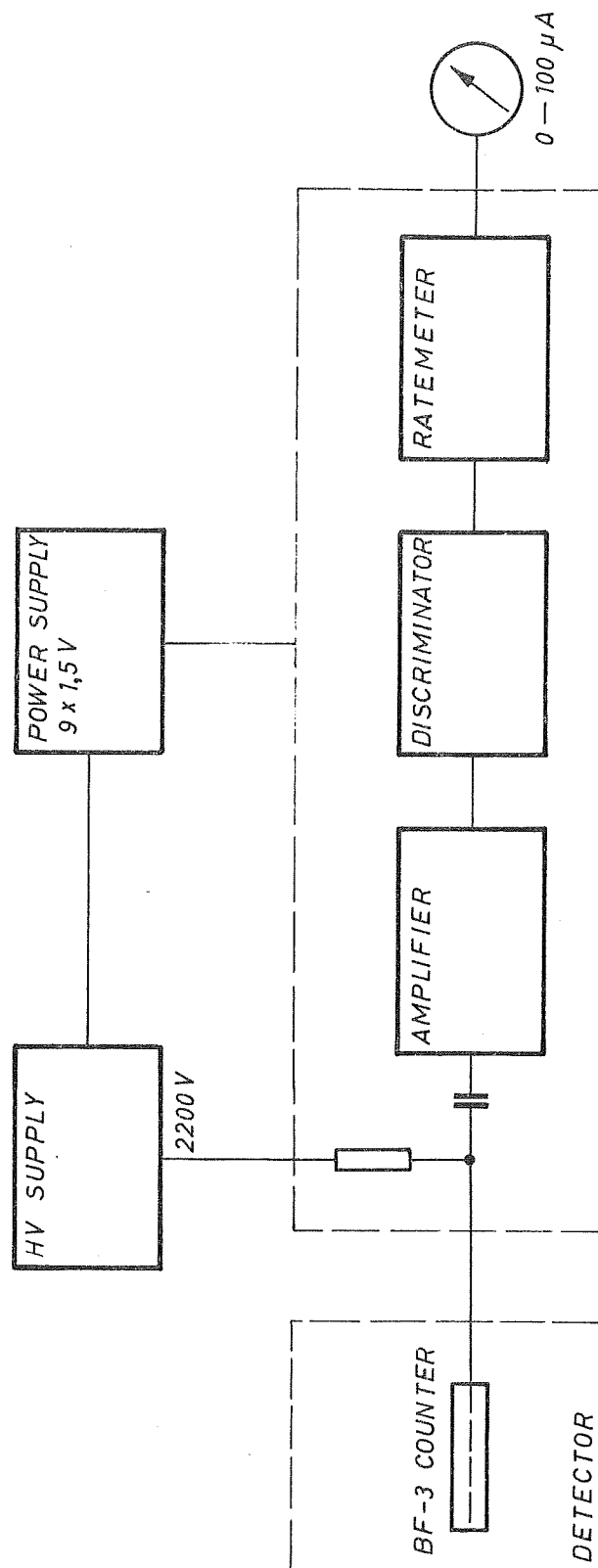


Fig. 8

Def.	Ant.	Benämning	Ritn.nr el. beteckn.	Material
NEUTRON DOSE RATE METER 2202 C,D Block diagram				Specifications and drawings subject to change without prior notice.
DI	Ritad 71-05-28 HG	Granskad GA	Godkänd	AB ATOMENERGI
Ersätter	Sammanst.	Beteckning	Skala	C-420300 B